

Demo of My L^AT_EX Style v1.5.2

Hassium

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This style uses PdfLaTeX editor. Slight changes would make the style compatible with XeLaTeX and LuaLaTeX. If you are new to L^AT_EX, I strongly recommend you to visit Overleaf for the basics. If you have any question, please contact me via discord: hassiummm

Special thanks to FSG and Bryce. Their advice on this style has been invaluable.

1 Packages

This style contains the following packages:

```
\usepackage[T1]{fontenc}
\usepackage[explicit]{titlesec}
\usepackage[utf8]{inputenc}
\usepackage{amsmath,amsthm,amssymb,amsfonts,mathrsfs,mathtools,nicematrix,chngecntr,
centernot,ytableau,sansmath,tikz-cd,pgfplots,tikz-3dplot,dutchcal,mathpazo,
spectralsequences,mathalpha}
\usepackage{imakeidx,textcomp,tocloft,environ,setspace,geometry,enumerate,enumitem,
blindtext,multicol,xcolor,fancyhdr,calligra,graphicx,etoolbox,wrapfig,pgfplots,
mdframed,tabularx,lipsum,comment,csquotes,verbatim,transparent,scalerel,ragged2e,
halloweenmath,manfnt,relsize,nameref,newpxtext}
\usepackage[wasysym]{horoscop}
\usepackage[hidelinks]{hyperref}
\usepackage{chemfig}
```

You can insert this style using the command “input”.

```
\documentclass{article}
\input{hassium.tex} % Insert the document by its path.
```

This style only uses very basic functions from the packages. If you are interested in writing complicated commands using them, please check their documentations on CTAN.

2 The Setup

After inserting the package, you should define the title and author name. The command “hsetup” gives you the title and the author name. The command “htoc” gives a table of contents, which we will mention later. The command “hmain” defines the style of the mainmatter, which includes a (not) fancy header. The “hfauthor” variable is the left part of the header. Here is an example, which is the code of this demo.

```
\documentclass{article}
\input{hassium.tex}
\begin{document}
  \def\htitle{Demo of My \LaTeX Style}
  \def\hauthor{Hassium}
  \def\hfauthor{Hassium}
  \hsetup
  \htoc
  \hmain
\end{document}
```

There are commands that adjust the margin of the document. Here “setstretch” adjusts the line spacing.

```
\geometry{letterpaper,top=60pt,bottom=60pt,left=60pt,right=60pt,headheight=12pt,
headsep=1.25\baselineskip}
\setstretch{1.25}
```

3 More on Table of Contents

You can add descriptions to each section and the description will appear in the table of contents, directly below the section name:

```
\section{This is a Sample Section}
\descr{This is a description to the section}
```

The table of contents only shows the section names, but no subsections and numberless sections. If you want a numberless section in the table of contents, use the “newsection” command:

```
\newsection{This is a numberless section}
```

There is a customized part in the table of contents. You can use the command “hpart” to define the name of a part. The numbering of parts are Roman numerals by default. An example is shown below:

```
\hpart{This is the first part}
\section{Groups}
\section{Rings}
\hpart{This is the second part}
\section{Manifolds}
```

Note that the section names and part names in the table of contents are hyperlinks; click on any section name to navigate directly to that section. You can do the converse to navigate to the table of contents as well.

4 Index Page and Reference Page

This style has a customized index page. The command “hdef” mark the word and print it with a fancy font. The command “hindex” is a customized index page that print words in three columns. Each page number in the index page contains a hyperlink to that page.

```
This is a \hdef{defintiion}. This is another \hdef{vocabulary}.
\hindex
```

You can also include a reference page using the command “hreference”. The first parameter is the style of reference and the second parameter is the name of your “.bib” document.

```
\hreference{amsalpha}{reference}
```

5 Other Environments and Commands

The line-spacing in “enumerate” environment is changed:

```
\setlist[enumerate]{topsep=0pt,itemsep=-1ex,partopsep=1ex,parsep=1ex}
```

The “level” environment is used in “enumerate” environment, consider the following code:

```
\begin{enumerate}
  \item This is the first line.
  \begin{level}
    \item This is the second line.
  \begin{level}
```

```

\item This is the third line.
\end{level}
\item This is another line.
\end{level}
\end{enumerate}

```

This code gives:

1. This is the first line.
2. This is the second line.
3. This is the third line.
4. This is another line.

The command “circled” draws a small circle and you can add something inside the circle:

```
\circled{1}
```

The output is ①. You can write any Roman numerals by:

```
\rom108
```

There are two simple commands for hand-written fonts:

```

\cfd{font 1}
\cfc{font 2}

```

The outputs are *font 1* and *font 2*.

6 Quiver

Quiver is done by varkor and AndréC, check their github for more information. I include quiver to draw curve arrows in a commutative diagram. To draw a diagram with quiver, check this website.

7 Theorem Styles

Several environments are offered:

```

\theoremstyle{definition}
\newtheorem{baseDefinition}{Definition}[section]
\newtheorem{baseTheorem}{Theorem}[section]
\newtheorem{baseProposition}{Proposition}[section]
\newtheorem{exercise}{Exercise}[section]
\newtheorem*{lemma}{Lemma}
\newtheorem*{corollary}{Corollary}
\newtheorem*{remark}{Remark}
\newtheorem*{notation}{Notation}
\newtheorem*{example}{Example}
\newtheorem{problem}{Problem}[section]
\newtheorem*{claim}{Claim}

```

The environment name can be customized by

```
\customtheorem{This is a custom theorem}
\begin{This is a custom theorem}
  The proof is trivial.
\end{This is a custom theorem}
```

The output will be:

This is a custom theorem. The proof is trivial.

If you are writing an appendix section, please use “`adefinition`” and “`atheorem`”. The numbering of these environments looks like “Definition/Theorem A1”.

```
\begin{adefinition}
  A definition in appendix.
\end{adefinition}
\begin{atheorem}
  A theorem in appendix.
\end{atheorem}
```

If you need a counter that is continued even if the section is changing, please use “`sdefinition`”, “`stheorem`”, and “`sproposition`”.

```
\begin{sdefinition}
  A definition.
\end{sdefinition}
\begin{stheorem}
  A theorem.
\end{stheorem}
\begin{sproposition}
  A proposition.
\end{sproposition}
```

If you need an exercise environment with numbering, use the environment `exercise*`. An example is shown below.

```
\begin{exercise*}[8.6]
  Prove the Poincaré conjecture.
\end{exercise*}
```

The result looks like:

Exercise 8.6. Prove the Poincaré conjecture. There is an automatic index in “`definition`”, “`theorem`”, and “`proposition`”. You can use the command “`quotethm`” to create a hyperlink to the theorem you quoted.

```
\begin{theorem}[Poincaré conjecture] % suppose it is theorem 1.10
  This is the Poincaré conjecture.
\end{theorem}
The proof of the Poincaré conjecture is here \quotethm{1.10}
```

Similarly, there are “`quotedef`” and “`quoteprop`”.

8 Invisible Proofs

The environment “reviewmode” is originally done by my friend ETwilight. It replaces your “proof” environment by three empty lines:

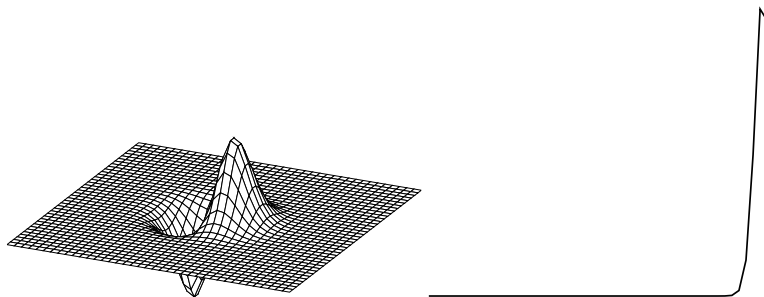
```
\begin{reviewmode}
  \begin{proof}
    The proof is trivial.
  \end{proof}
\end{reviewmode}
```

9 Drawing Commands

The function drawing command is based on the package pgfplots, you may use the following code:

```
\functiondraw{3}{0.8}{-5}{5}{\exp(-x^2-y^2)*x}
\functiondraw{2}{0.8}{-5}{5}{\exp(x^2-2*\sin(deg(x)))}
```

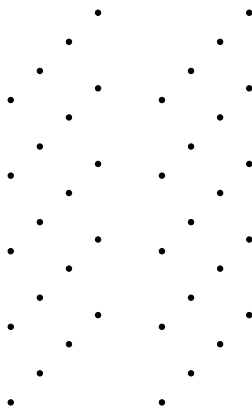
The first parameter decides the dimension of the graph. If you want the graph to be 2D, then set 2 to the first parameter. Otherwise, give a random integer that does not equal to 2 would give you a 3D graph. The second parameter is the scale of your plot. The third and the fourth is the domain of the plot, in our example, the domain is $[-5, 5]$. The last entry is for your function. The example code (with centering) yields the following plot.



You might need some points that lie uniformly. This can be simply done by the “points” command.

```
\begin{tikzpicture}
  \points{5}{2}{4}{1}{2}{1}
\end{tikzpicture}
```

Here the first three parameters are the number of points in x, y, z -axes. The last three parameters are distances of points in each axis. This command is helpful when you draw Cayley graphs.



10 Simple Commands in Math Mode

I will give a table of all commands in math mode.

<code>\ua</code>	$\uparrow$	<code>\colim</code>	colim
<code>\da</code>	$\downarrow$	<code>\coker</code>	coker
<code>\Ra</code>	$\Rightarrow$	<code>\cat{C_S}</code>	$\mathcal{C}_S$
<code>\La</code>	$\Leftarrow$	<code>\Cl</code>	Cl
<code>\Ua</code>	$\Uparrow$	<code>\CAT</code>	CAT
<code>\Da</code>	$\Downarrow$	<code>\card{1}</code>	$ 1 $
<code>\nRa</code>	$\nRightarrow$	<code>\hash</code>	$\#$
<code>\nLa</code>	$\nLeftarrow$	<code>\dist</code>	dist
<code>\hra</code>	$\hookrightarrow$	<code>\diag</code>	diag
<code>\hla</code>	$\hookleftarrow$	<code>\Div</code>	Div
<code>\lt</code>	$\sim$	<code>\Cl</code>	Cl
<code>\mt</code>	$\mapsto$	<code>\sq</code>	$\square$
<code>\rat</code>	$\rightharpoonup$	<code>\defa</code>	$\coloneqq$
<code>\lat</code>	$\leftarrow$	<code>\pa</code>	∂
<code>\thra</code>	$\rightarrow$	<code>\d</code>	d
<code>\thla</code>	$\leftarrow$	<code>\Ext</code>	Ext
<code>\bij</code>	$\xrightarrow{\sim}$	<code>\Tor</code>	Tor
<code>\Alt</code>	Alt	<code>\ev</code>	ev
<code>\Ann</code>	Ann	<code>\fl{1}</code>	$[1]$
<code>\A</code>	$\mathbb{A}$	<code>\Frac</code>	Frac
<code>\ab</code>	α	<code>\al</code>	α
<code>\bnc{1}{2}</code>	$\binom{1}{2}$	<code>\be</code>	β
<code>\bcs</code>	$\beth$	<code>\ga</code>	γ
<code>\Cay</code>	Cay	<code>\de</code>	δ
<code>\bb{H}</code>	$\mathbb{H}$	<code>\ep</code>	ϵ
<code>\ca{H}</code>	$\mathcal{H}$	<code>\si</code>	σ
<code>\fr{H}</code>	$\mathfrak{H}$	<code>\la</code>	λ
<code>\scr{H}</code>	$\mathscr{H}$	<code>\ka</code>	κ
<code>\comp</code>	$\circ$	<code>\om</code>	ω
<code>\iso</code>	$\approx$	<code>\Ga</code>	Γ
<code>\niso</code>	$\not\approx$	<code>\De</code>	Δ
<code>\Mor</code>	Mor	<code>\Si</code>	Σ
<code>\Aut</code>	Aut	<code>\LA</code>	Λ
<code>\End</code>	End	<code>\Om</code>	Ω
<code>\Hom</code>	Hom	<code>\vt</code>	ϑ
<code>\Inn</code>	Inn	<code>\vp</code>	φ
<code>\Out</code>	Out	<code>\ve</code>	ε
<code>\Iso</code>	Iso	<code>\acts</code>	$\curvearrowright$
<code>\Ob</code>	Ob	<code>\Gal</code>	Gal
<code>\dom</code>	dom	<code>\cyc{1}</code>	$\langle 1 \rangle$
<code>\cod</code>	cod	<code>\Ht</code>	ht

$\backslash wh\{1\}$	$\hat{1}$	$\backslash supe$	$\supseteq$
$\backslash Hol$	Hol	$\backslash nsub$	$\not\subset$
$\backslash id$	id	$\backslash nsup$	$\not\supset$
$\backslash im$	im	$\backslash nsube$	$\not\subseteq$
$\backslash inv\{1\}$	1^{-1}	$\backslash nsupe$	$\not\supseteq$
$\backslash Int$	Int	$\backslash subn$	$\subsetneq$
$\backslash eqc$	$\neq$	$\backslash supn$	$\supsetneq$
$x \backslash mod y$	$x \bmod y$	$\backslash es$	$\emptyset$
$\backslash norm\{1\}$	$\ 1\ $	$\backslash sm$	$\backslash$
$\backslash N$	$\mathbb{N}$	$\backslash ps$	$\mathcal{P}$
$\backslash C$	$\mathbb{C}$	$\backslash Un$	$\cup$
$\backslash R$	$\mathbb{R}$	$\backslash In$	$\cap$
$\backslash Q$	$\mathbb{Q}$	$\backslash Du$	$\sqcup$
$\backslash Z$	$\mathbb{Z}$	$\backslash Cp$	$\coprod$
$\backslash F$	$\mathbb{F}$	$\backslash cp$	$\amalg$
$\backslash nsg$	$\leq$	$\backslash sgn$	sgn
$\backslash ot$	$\otimes$	$\backslash sdp$	$\times$
$\backslash op$	$\oplus$	$\backslash Spec$	Spec
$\backslash odd$	odd	$\backslash Syl$	Syl
$\backslash even$	even	$\backslash Sym$	Sym
$\backslash ord$	ord	$\backslash Mod$	Mod
$\backslash Ps\{1\}$	$\mathbb{P}^1$	$\backslash Sg$	$\mathfrak{S}$
$\backslash CP\{1\}$	$\mathbb{CP}^1$	$\backslash Ag$	$\mathfrak{A}$
$\backslash RP\{1\}$	$\mathbb{RP}^1$	$\backslash uni$	$\exists!$
$\backslash proj$	proj	$\backslash tp\{1\}$	$1^\top$
$\backslash po$	$\prec$	$\backslash T$	$\mathcal{T}$
$\backslash poe$	$\preceq$	$\backslash tri$	$\triangle$
$\backslash ran$	ran	$\backslash td$	tradg
$\backslash sub$	$\subset$	$\backslash wt\{1\}$	$\tilde{1}$
$\backslash sube$	$\subseteq$	$\backslash wb\{1\}$	$\bar{1}$
$\backslash sups$	$\supset$	$\backslash Vee$	$\vee$